

Ricardo Vicente

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Motivated Computer Science and Engineering student at Instituto Superior Técnico with a strong foundation in algorithms, data structures, computer architecture, and systems programming. Co-developed a mobile-first campus room finder backend in Django, programmed an AI-driven application in Python, implemented custom data structures in C, and built a Mini-Transformer entirely in RISC-V Assembly. Interested in technology consulting, full-stack development, and solving technically challenging problems through efficient and reliable software.

Education

BSc in Computer Science and Engineering

Instituto Superior Técnico, Lisbon, Sep 2025 – Present

Current GPA: 16.5 / 20

Academic Highlights: Logic for Programming (20/20), Programming Fundamentals (18/20), Computer Architecture (18/20)

Relevant coursework: Algorithms and Data Structures, Calculus, Linear Algebra, Physics, Management

Projects

Spotter – Campus Room Finder

Instituto Superior Técnico | Team Project, Summer 2026

- Built the Django and PostgreSQL backend for a mobile-first app tracking campus room availability.
- Designed recursive ETL pipelines to fetch and normalize space and schedule data via the Fénix API.
- Implemented an idempotent DB sync engine with atomic upserts, stale data removal, and unit tests.

Billing System in C

Instituto Superior Técnico | Year 1, Semester 2

- Built a terminal-based billing and inventory system in C to manage products, shopping carts, and invoices.
- Implemented custom Hash Tables and Linked Lists from scratch for high-performance data lookups.
- Ensured strict memory management with zero leaks, fully tested using Valgrind.

Rule-Based Scrabble AI in Python

Instituto Superior Técnico | Year 1, Semester 1

- Developed a text-based Scrabble game in Python featuring an AI opponent with three distinct difficulty levels.
- Implemented word validation and pattern-matching algorithms using a 25,000-word vocabulary list.
- Structured the codebase using Abstract Data Types (ADTs) and custom randomization algorithms (such as Fisher–Yates).

RISC-V Mini-Transformer

Instituto Superior Técnico | Year 1, Semester 2

- Built a Mini-Transformer model in 32-bit RISC-V Assembly to predict the next word in a sentence.
- Programmed core attention mechanisms from scratch, including Q/K/V matrices and dot-product scoring.
- Developed low-level system routines for file parsing, memory management, and manual overflow detection.

Skills

Programming: Python, C, Prolog, RISC-V Assembly

Frameworks & Tools: Django, REST APIs, Git, Linux/WSL, VS Code

Others: Structured Problem-Solving, Teamwork, Analytical Thinking, Autonomy & Time Management

Languages

Portuguese – Native | **English** – Professional working proficiency

Interests

Technology, Entrepreneurship, Football, Music